Project FORESIGHT – Demand & Inventory Intelligence

Project Name: Project FORESIGHT
Report Type: Business Insights Report
Prepared By: Utkarsh Chaudhari
Tools Used: Python, Pandas, Matplotlib, Streamlit, Machine Learning

1. Introduction

This report summarizes the key business insights generated from historical sales, product information, inventory data, and demand forecasting. The objective is to help business managers make informed decisions related to inventory planning, sales optimization, and product management.

2. Top Selling Products

The analysis identifies the products with the highest sales volume.

Rank	Product Name	Category	Units Sold
1	Office Chair	Furniture	12,450
2	Study Table	Furniture	11,920
3	Bookshelf	Furniture	10,840
4	Gaming Chair	Furniture	9,760
5	Computer Desk	Furniture	9,140

Business Insight

- Furniture products generate the highest sales.
- Office Chairs consistently remain the best-selling product.
- High-demand products require continuous inventory monitoring.

Recommendation

- Maintain higher safety stock for top-selling products.
- Prioritize these products during procurement and warehouse allocation.

3. Low Selling Products

Products with consistently low sales are listed below.

Rank	Product Name	Category	Units Sold
1	Desk Lamp	Lighting	2,150
2	Wall Shelf	Furniture	2,480
3	Storage Box	Accessories	2,910
4	Table Clock	Accessories	3,140
5	Pen Holder	Accessories	3,420

Business Insight

- Several products show low customer demand.
- These items occupy warehouse space without contributing significantly to revenue.

Recommendation

- Offer discounts or promotional bundles.
- Reduce future procurement quantities.
- Review whether low-performing products should be discontinued.

4. Monthly Sales Trend

Month	Sales Trend
January	Medium
February	Medium
March	High
April	High
May	Medium
June	Medium
July	High
August	High
September	Medium
October	Very High
November	Very High
December	Highest

Observation

- Sales increase during festive seasons and year-end periods.
- October to December contributes the largest share of annual revenue.

Business Recommendation

- Increase inventory before high-demand months.
- Schedule marketing campaigns during peak seasons.
- Plan supplier orders in advance to avoid stock shortages.

5. Category Performance

Category	Revenue Contribution	Performance
Furniture	48%	Excellent
Electronics	28%	Good
Lighting	15%	Average
Accessories	9%	Needs Improvement

Business Insight

- Furniture is the strongest-performing category.
- Accessories contribute the least revenue.

Recommendation

- Expand the Furniture product range.
- Improve marketing for Accessories.
- Evaluate pricing strategies for low-performing categories.

6. Inventory Summary

Metric	Value
Total Products	200
Average Inventory	115 Units
Products Below Reorder Level	28

Overstocked Products	15
Healthy Inventory	157

Business Insight

- Most products maintain healthy inventory levels.
- A subset of products faces stockout risk.
- Overstocked products increase storage costs.

Recommendation

- Reorder products below their reorder point immediately.
- Introduce promotions for overstocked items.
- Review reorder thresholds periodically.

7. Risk Scoring

Risk scoring is performed at the **SKU level** to identify stockout and overstock risks and support inventory decision-making.

a. Stockout Risk Score

Stockout risk is calculated based on **Days of Supply (DOS)**:

Days of Supply	Stockout Risk Score	Risk
≤ 3 days	1.00	HIGH
≤ 7 days	0.75	HIGH
≤ 14 days	0.40	MEDIUM
> 14 days	0.10	LOW

b. Overstock Risk Score

Overstock risk is calculated based on **Days of Supply (DOS)**:

Days of Supply	Overstock Risk Score	Risk
≥ 90 days	1.00	HIGH
≥ 60 days	0.75	HIGH
≥ 30 days	0.40	MEDIUM
< 30 days	0.10	LOW

c. **Capital Locked**

Capital locked in inventory is calculated as:

Capital Locked = Current Inventory × Average Unit Price

This represents the estimated value of capital currently tied up in inventory.

d. **Sales at Risk**

Potential sales at risk due to stockout are calculated as:

Sales at Risk = Average Daily Demand × Average Unit Price × 7 × Stockout Risk Score

This estimates the potential revenue exposure over a 7-day period.

e. **Risk Level**

The overall SKU risk level is determined using the stockout and overstock risk scores:

- **HIGH** → Immediate inventory attention required.
- **MEDIUM** → Monitor and plan corrective action.
- **LOW** → Normal inventory monitoring.

f. **Primary Risk**

Each SKU is assigned a primary risk:

- **STOCKOUT** → Stock availability is insufficient relative to demand.
- **OVERSTOCK** → Inventory exceeds expected demand significantly.
- **BALANCED** → Inventory is within an acceptable range.

g. **Recommended Action**

Recommended actions are generated based on the identified risk:

- **HIGH STOCKOUT** → Urgently replenish stock.
- **MEDIUM STOCKOUT** → Plan replenishment.
- **HIGH OVERSTOCK** → Reduce inventory / run promotions.
- **MEDIUM OVERSTOCK** → Monitor inventory and reduce future procurement.
- **BALANCED** → Maintain current inventory level.

h. **SKU-Level Risk Output**

For every SKU, the risk analysis CSV contains:

1. **Stockout Risk Score**

2. **Overstock Risk Score**
3. **Risk Level**
4. **Primary Risk**
5. **Recommended Action**
6. **Sales at Risk**
7. **Capital Locked**

This provides a clear, measurable, and actionable risk assessment for inventory management.

8. Risk-Based Business Insights

Metric	Value
Total SKUs Analysed	200
Total ₹ Sales at Risk	₹26,844,927.09
Total ₹ Capital Locked	₹4,587,583.00
High Risk SKUs	190
Medium Risk SKUs	10
Low Risk SKUs	0
Stockout Risk SKUs	200
Overstock Risk SKUs	0
Balanced SKUs	0

Top Risk Insight:

- 190 SKUs are classified as **HIGH risk**.
- Stockout is the **primary risk for all 200 SKUs**.
- Approximately **₹26.84 million sales are at risk**.
- Approximately **₹4.59 million capital is locked in inventory**.
- High-risk SKUs should be prioritized for replenishment.

9. Key Business KPIs

KPI	Value
Total Sales	73,000 Units
Average Daily Sales	200 Units
Best Selling Category	Furniture
Peak Sales Month	December
Average Inventory	115 Units

Forecast Accuracy (R^2)	0.94

10. Strategic Recommendations

a. Sales Strategy

• Focus marketing on high-performing categories.
• Launch promotions during seasonal demand peaks.
• Use historical demand for campaign planning.

b. Inventory Strategy

• Maintain safety stock for high-demand SKUs.
• Monitor inventory daily using automated alerts.
• Reduce overstock through targeted promotions.
• **Prioritize high-risk SKUs based on Stockout Risk Score and Sales at Risk.**
• **Monitor ₹ Sales at Risk and ₹ Capital Locked to support inventory decisions.**

c. Product Strategy

• Expand high-performing product lines.
• Improve visibility of slow-moving products.
• Periodically review product portfolio performance.

d. Forecasting Strategy

• Retrain the forecasting model regularly using the latest sales data.
• Incorporate external factors such as holidays and promotional events into future model versions.

11. Conclusion

The business insights reveal clear sales patterns, product performance trends, and inventory opportunities. High-performing products should receive priority in procurement and inventory planning, while slow-moving products require targeted promotional strategies.

By using these insights alongside the demand forecasting model, organizations can:

- Improve inventory utilization.
- Reduce stockout and overstock risks.
- Increase customer satisfaction.
- Support data-driven business decisions.
- Enhance overall operational efficiency.